

Alexander Loker

aloker@uwaterloo.ca | linkedin.com/in/alexander-loker | alexloker.com

Education

University of Waterloo

2025 – Present

Bachelor of Applied Science in Mechatronics Engineering

Skills

Hardware & CAD: SolidWorks, Altium, LtSpice, AutoCAD, Raspi, Soldering, 3D Printing

Software: C++, Python, Git, Linux

Firmware: PWM, DMX, I2C, SPI, UART, PID, Raspi

Experience

WET Design

May – August 2026

Product Designer & Engineer

Sun Valley, CA, USA

- Designed a **DMX** lighting tester ground up to solve price issues, introduce voltage-safety, and allow script injection
- Validated the design for a lighting tester and a new product in parallel, spanning electronics, mechanical, software, and fluids fabrication, with SolidWorks designed stands to mount each mock-up device.
- Collaborated with team members on a new Electrical Box, using tools like Draw.io to communicate the wiring diagram, SolidWorks to understand space required, and organizing meetings to review progress.

Projects

DMX Handheld Light Tester – WET Design

June – August 2026

- Developed a fully custom **Raspi CM4** carrier board, BMS, and user interface alongside a company RS-485 port in **Altium** from part selection, schematic, to full PCB layout.
- Maintained **SI** and reduced **EMI** through guarding differential pairs, a ferrite-bead pi filter, tight local **decoupling** at each IC's power pins, and a minimized switching-current loop to reduce parasitic inductance.
- Integrated battery management and power distribution, implementing thermal balancing with low-impedance routing across a **BQ charger IC**, **buck-boost IC**, and CM4 3.3V rail to safely power the board.
- Wrote the tester's firmware in **Python**, implementing **UART**, **I2C**, and **SPI** protocol logic to communicate with the DMX fixtures, BMS, and LCD Screen respectively.
- Created the physical UI in **SolidWorks** and **PyGame**, ensuring holdability, reliability, and an aesthetically pleasing look via precise fits, tolerance, and feedback iteration, alongside a readable and informative interface.

Companion Robot

January – April 2026

- Led end-to-end design of a companion robot integrating mechanical design (**SolidWorks**), electrical components and routing, and firmware implementation (**Python**).
- Designed **hardware and firmware** for a tri-omniwheel drivetrain on a **Pi Pico** with **Python PID** motor control and an open-source tracking algorithm, enabling **autonomous 360°** human-following
- Compacted robot volume by **26%** by redesigning the mechanical housing alongside embedding servos, microphones, and a camera module directly into the shell, utilizing compact interlocking joints

Under-voltage Lockout Circuit – Biomechatronics Exoskeleton Team

May – August 2026

- Designed and verified an UVLO circuit schematic in **LtSpice**, and transformed the schematic into a PCB in **Altium**, confirming safe voltage input to sensitive components.
- Created under-voltage lockout logic via **Schmitt trigger hysteresis** and controlled current/voltage flow through a **dual-MOSFET switch system** ensuring voltage lockout at **24.1V**

Autonomous Air Hockey Defense Machine

November 2025

- Designed and assembled a **fully automated** puck-deflection robot using **one distance sensor**, achieving a **76%** interception rate against live opponents
- Integrated an **optical sensor** for real-time foul detection and a **distance sensor** for trigonometric puck trajectory prediction, collecting and analyzing sensor data to validate system accuracy
- Developed a real-time **predictive algorithm in C++** to calculate puck landing position from sensor input.